Maintenance Treatment of Schizophrenic Outpatients With a Depot Phenothiazine

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■ An increasing number of outpatients are receiving psychotropic drugs as part of the treatment prescribed by both general practitioners and specialists. It is a common experience to find that chronic schizophrenic patients, once released from the hospital, terminate their medication.[12,20] If no immediate behavioral deterioration occurs, the patient or his relatives usually conclude that he no longer requires the drug. Although true for some patients, clinical experience and controlled studies have shown that the majority continue to require maintenance treatment.[4-6,11]

One of the main reasons for discontinuation of treatment is the inconvenience of taking medication several times a day. For the patients requiring maintenance treatment, the availability of a compound with a prolonged duration of action has been needed for some time. In recent years fluphenazine, a major psychotropic drug, has become available in an enanthate form, which is a heptanoic acid ester of fluphenazine, and a decanoate form, which is a decanoic acid ester of fluphenazine. These compounds, given as a single dose intramuscularly, are reported to be effective for a period of about two weeks.[1-3,9,10,13-17,19]

Fluphenazine decanoate has been reported to have fewer side reactions and more prolonged effect than the enanthate form.[19] In a previous study, we found fluphenazine decanoate* to be a very effective psychotropic compound without producing severe side effects in the treatment of an inpatient population.[8]

The aims of this study were to answer the following questions: 1. Do stabilized chronic schizophrenic outpatients need to continue their prescribed medications after being an outpatient more than an average of 18 months? 2. Can fluphenazine decanoate, administered parenterally at fixed intervals enable these chronic schizophrenic outpatients to maintain their improved condition?

MATERIAL AND METHOD

Twenty-four chronic schizophrenic outpatients whose population characteristics are shown on Table I were included in this study. There were 20 female and four male patients ranging in age from 22 to 50 years with an average of 36 years. Diagnostic subgroup distribution was 12 paranoid, 3 hebephrenic, 2 catatonic, 1 simple and 6 chronic undifferentiated. The duration of the illness since first psychiatric hospitalization ranged from 5 to 25 years, averaging 12.4 years. The outpatient status period ranged from six months to three years three months, with an average of 1.5 years. The patients had been receiving drug treatment for six months to three years six months, with an average of two years for the group. None of the patients had any evidence of organic brain syndrome or systemic disease. They have been followed up regularly as a part of the Therapy Resistant Psychosis Project while maintained on a piperazine phenothiazine.

To establish a baseline, doctor's choice medications were discontinued gradually over a one-week period and replaced by oral daily fluphenazine hydrochloride. Based on our previous experience, the dosage ratio was considered to be 2 mg butaperazine to 1 mg fluphenazine hydrochloride, and 1 mg to 1 mg for other piperazine phenothiazines. After a

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TABLE I. POPULATION CHARACTERISTICS (N.24)

	Age	Duration of Illness (since 1st hosp.)	Duration of Outpatient Status	Duration of Dr.'s Choice Treatment	Diagnosis: Schizophrenia Paranoid	Hebephrenic	Catatonic	Simple	Chr. Undiff.
Average (yrs.)	36	12.4	1.5	2	12	3	2	1	6
Range (yrs.)	22-50	5-25	0.6-3.3	0.6-3.6					

minimum of six weeks oral fluphenazine treatment, the patients were randomly assigned into two groups. Group "A" of 13 patients was to receive fluphenazine decanoate (25 mg per ml) by intramuscular injection, and group "B" of 11 patients was to receive a placebo injection of sesame oil. Both the active compound and placebo were administered bimonthly as a single injection. The injected mg amount was approximately equal to the patient's daily oral fluphenazine dosage. If deterioration had occurred by the next due injection, placebo was replaced by fluphenazine decanoate 2.5 mg I.M. If deterioration continued, dosage and injection intervals were thereafter increased by the investigator as he saw clinically fit.

The patients receiving an antiparkinson medication were maintained on it throughout all phases of the study including doctor's choice, fluphenazine hydrochloride, fluphenazine decanoate or placebo treatment.

Every patient attended the outpatient department every two weeks for clinical assessment during the six months study period.

A seven point rating scale was used monthly for the global clinical evaluation, which ranged from 1, indicating maximal improvement with no overt sign of psychosis, to 7, indicating severe psychosis. Psychopathological and drug-induced psychosomatic changes were evaluated every two weeks using our rating scales (Itil-Keskiner),[7] of 70 and 40 items respectively. The ratings were completed by the same psychiatrist throughout the study period.

Laboratory examinations were carried out at regular intervals for CBC, alkaline phosphatase, SGOT, BUN, bilirubin, thymol turbidity and urinalysis. EKG's and EEG's were performed monthly.

RESULTS

The 24 patients completed six weeks daily oral fluphenazine treatment and 12 weeks of bimonthly injections of fluphenazine decanoate or placebo treatment.

Distribution of the dosage and number of patients of the parenteral treatment period is shown on Table II. Group "A" patients received parenteral injections regularly every second week. The increase in the total and average amount of medication noticed at the eighth week was due to the adjustment of dosage in three patients showing signs of relapse. Of these, one patient required readmission. The single bimonthly parenteral dose ranged from 12.5 mg to 75 mg. for this group.

Of the 11 patients in group "B", two manifested signs of relapse within the first week of the placebo period and required immediate changeover to active compound. By the time the second injection was due, four patients required active drug, and by the sixth week, eight had been started on fluphenazine decanoate. Despite increased dosage and shortening of the injection periods, two patients had to be readmitted; one temporarily at the second week and one at the fourth week of the parenteral treatment period. Only three patients maintained their improved condition throughout the study period while remaining on bimonthly placebo injections. Because of recurrence of psychotic symptomatology in the placebo group, the total and average dosage of fluphenazine decanoate required by group "B" patients was much higher than of group "A" patients (Table II). The effective single parenteral dose of fluphenazine decanoate in group "B" ranged from 2.5 mg to 75 mg.

The dosage of antiparkinson medication (procyclidine) remained much the same throughout the whole study period (Table

TABLE II DISTRIBUTION OF THE DOSAGE AND PATIENTS OF THE PARENTERAL TREATMENT PERIOD

	Medication		Start	Week of Parenteral Treatment 1	2	4	6	8	10	12
Group "A" (N 13)	I.M Flu.Dec. Dose (mg.)	No. of Pts.	13		13	13	13	13	13	13
		Total	362.5		362.5	362.5	362.5	377.5	390	415
		Average	28		28	28	28	29	30	32
Group "B" (N 11)	I.M Placebo Dose (mg.)	No. of Pts.	11		7	4	3	3	3	3
		Total	–		–	–	–	–	–	–
		Average	–		–	–	–	–	–	–
	I.M. Flu.Dec. Dose (mg.)	No. of Pts.	0	2	4	7	8	8	8	8
		Total	0	5	67.5	160	235	257.5	270	307.5
		Average	0	2.5	17	23	29	32	34	38

TABLE III VARIATION IN ANTIPARKINSON MEDICATION (PROCYCLIDINE) DURING THE THREE TREATMENT PERIODS

		Dr.'s Choice	Daily Oral Flu. Hcl.	Bimonthly Parenteral Treatment Period (Flu. Dec. or Placebo)
Group A (N 13)	Number of pts.	(9)	(9)	(10)
	Average Daily Dose	9.5 mg.	8.8 mg.	8 mg.
Group B (N 11)	Number of pts.	(8)	(8)	(8)
	Average Daily Dose	8.7 mg.	8 mg.	8.7 mg.

III). There was no need for increases in average dose or in the number of patients who required antiparkinson medication.

Global Evaluation.—The last global behavior ratings of oral fluphenazine treatment were compared with the ratings at the second, fourth, eighth and twelfth week of parenteral treatment (fluphenazine decanoate or placebo). Changes in global score with parenteral treatment is shown in Table IV. None of the patients were scored more than "4" at any treatment period. The score distribution of global ratings during oral fluphenazine treatment for groups "A" and "B" were similar. In group "A" patients there were no significant changes in the ratings of bimonthly fluphenazine decanoate compared with the last ratings of the daily oral fluphenazine treatment period. On the other hand, group "B" patients showed significant changes in global behavior at second week and fourth week of the parenteral placebo treatment when compared with the last ratings of oral fluphenazine treatment. After eight of the eleven patients had been back on active compound, the changes in global score were no longer significant for this group.

Psychopathological Evaluation.—The analysis of changes in the 70 individual symptoms from oral to parenteral medication treatment

TABLE IV CHANGE IN GLOBAL SCORE WITH PARENTERAL (FLUPHENAZINE DECANOATE OR PLACEBO) TREATMENT

	Score	End of Oral Fluphenazine Period (N)	Parenteral Treatment Period (weeks) (I.M. Fluphenazine Decanoate)			
Group "A" (N=13)	1	6	6	6	5	6
	2	6	6	6	7	6
	3	1	1	1	–	–
	4				1	1
	5					
	6					
	7					
	Oral Fluphenazine vs. I M Fluphenazine Dec.		N.S*	N.S.	N.S.	N.S.
			(I.M. Fluphenazine Decanoate or Placebo)			
Group "B" (N=11)	1	5	3	2	3	3
	2	5	3	4	7	7
	3	1	5	5	–	–
	4				1	1
	5					
	6					
	7					
	Oral Fluphenazine vs. I.M. Medication (Placebo)		$P \leq 0.01$*	$P \leq 0.03$	N.S.	N.S.

*Sign Test

periods did not reveal statistically significant* differences in the single symptom of either group. However, the cumulative symptom analysis (Shapiro-Itil)[18] showed differences between group "A" and "B" and within the groups as well, at different treatment periods in the total symptomatology (Fig. 1). This analysis represents the average of the sum of scores for each of the 70 symptoms for the total population at various periods of the study.

During doctor's choice treatment period (Fig. 1, graph I), group "A" patients showed more cumulative average symptoms scores than group "B", indicating that the groups were not equally matched for psychopathology probably due to random assignment. During daily oral fluphenazine (Fig. 1, graph II) both groups showed a decrease in the total symptomatology, indicating that oral fluphenazine treatment was more effective in both populations than doctor's choice medication. This was particularly so for group "B" patients. This condition was reversed at the fourth week of the bimonthly parenteral treatment period (Fig. 1, graph III) where the majority of group "B" patients were receiving placebo injections, and the sum of psychotic symptoms increased above the oral treatment period. The increased slope seen in the affective cluster was due to recurrence of symptoms of anxiety, ambivalence and depression. The increase of symptomatology was also seen in the clusters of person in the form of autism and passivity, in volition with symptoms of disinterest and ambivalence, in delusions as the appearance of suspiciousness and ideas of reference, in motor behavior as restlessness and in higher mental functions mainly as disturbance of attention and concentration and in general attitude manifested mainly as withdrawal and irritability. During the same period, group "A" continued to show a decrease in the total symptomatology indicating further improvement. At the twelfth week of parenteral treatment (Fig. 1, graph IV) both group scores were similar, showing that the psychotic symptoms of group "B" which had recurred were controlled after majority of patients had been on fluphenazine decanoate.

Secondary effects.—There were no physical or neurological complications during this study and no abnormal laboratory deviations. The incidence of extrapyramidal symptoms was no different between the oral fluphena-

*Chi-square test.

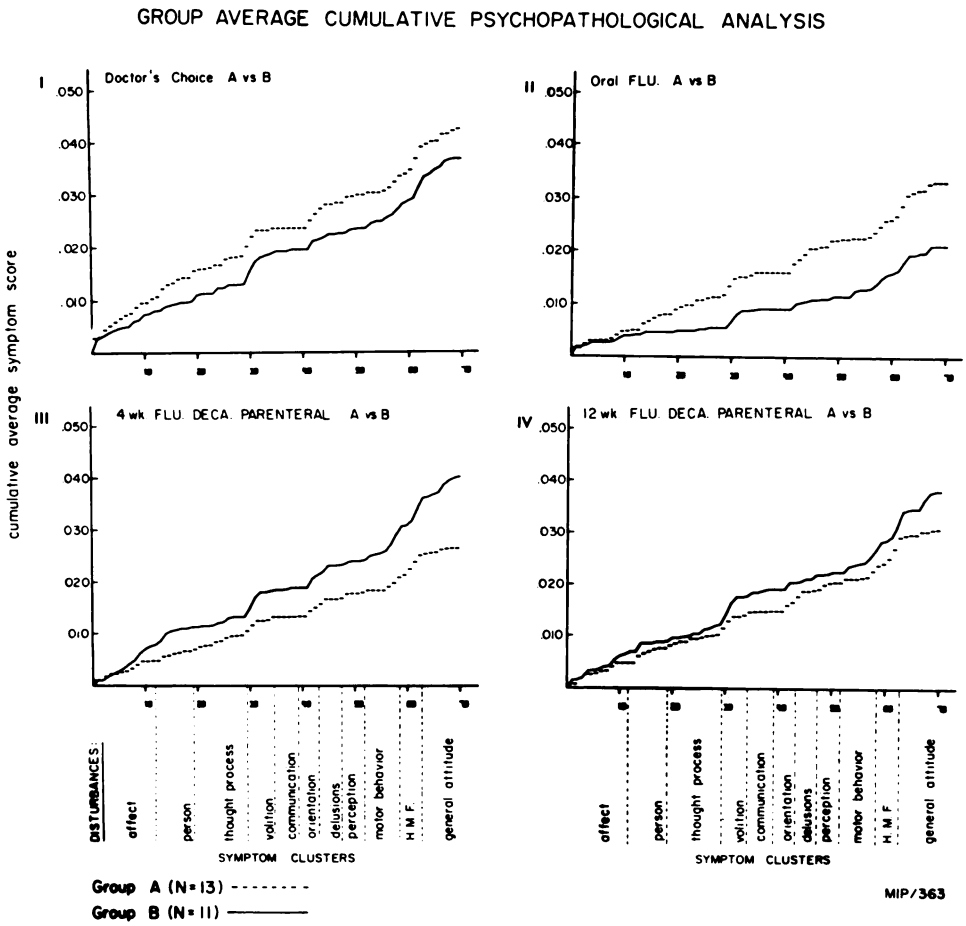


Figure 1.

zine and parenteral fluphenazine treatment periods.

DISCUSSION AND CONCLUSION

The results indicated that in chronic schizophrenic outpatients maintenance treatment is required for the majority of cases. Early signs of relapse in placebo treated patients were: withdrawal, disinterest and disturbance of attention and concentration.

Fluphenazine decanoate was found to be an effective compound in the control of psychotic symptoms when administered intramuscularly at bimonthly intervals and comparable to daily oral fluphenazine hydrochloride treatment. Individual maintenance dosage ranged from 2.5 mg to 75 mg.

There were no differences in the incidence of extrapyramidal symptoms between daily oral and parenteral long acting treatment.

The practical applications of fluphenazine decanoate or similar long acting psychotropic compounds are numerous and may be a solution to the problem of maintenance treatment in stabilized psychotic outpatients.

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Everyone is a prisoner of his own experiences. No one can eliminate prejudices—just recognize them.

—EDWARD R. MURROW